Run One sjSDM CV Calibration Budget Rung

Description

Fits and scores every regularization candidate over cached prepared folds for one or more repeats at one fixed iteration and sampling budget.

Usage

```
run_sjsdm_cv_calibration_rung(
  data_candidates = NULL,
  list_prepared_folds = NULL,
  data_budget = NULL,
  sel_abiotic_formula = NULL,
  config_sjsdm_cv_fitting = NULL,
  repeat_ids = 1L,
  fold_ids = NULL,
  work_checkpoint_file = NULL,
  calibration_stage = "calibration",
  seed = 900723L,
  device = "gpu",
  candidate_fit_function = fit_sjsdm_cross_validation_candidate,
  predict_function = predict_sjsdm_probability_matrix,
  score_function = score_sjsdm_joint_tuning_predictions,
  verbose = TRUE
)
```

Arguments

`data_candidates`	Complete regularization candidate table.
`list_prepared_folds`	Cached fold records from [prepare_sjsdm_tuning_folds()].

`data_budget`	One-row calibration ladder entry.
`sel_abiotic_formula`, `config_sjsdm_cv_fitting`	Model formula and validated structural CV fitting configuration.
`repeat_ids`	Integer repeats to execute.
`fold_ids`	Optional positive integer fold IDs to retain. 'NULL' keeps all folds.
`work_checkpoint_file`	Optional '.qs' file that persists each completed candidate-fold result.
`calibration_stage`	Character label used in progress messages and checkpoint provenance.
`seed`, `device`	Deterministic seed and fitting device.
`candidate_fit_function`, `predict_function`, `score_function`	Injectable fitting, prediction, and scoring backends.
`verbose`	Logical. If 'TRUE', progress is printed to the console.

Value

List containing fold-level benchmark evidence and attempt-level provenance.